

Grace Driskill

CSC 343 Spring 2022

May 3, 2022

Project: Connect 4

Requirements:

I plan on taking this project on from the angle of it eventually being a game machine in an arcade. This specific context will affect some aspects of the requirements. The game needs to be fully functional. It should indicate which player's turn it is, and then give them a way to choose where to place their piece. When someone gets four in a row, the game needs to alert the players that the game is over and who won. Then, the winning user should be prompted to enter their name to show up next to their score on the scoreboard.

My usability goals are that the game is easy to learn to play for a wide range of users (users should be able to fully understand how to play the game on their first time playing without needing outside help), and functionable (pieces go correctly where they were placed, game ends when there are four pieces in a row). My user experience goals are that the game is fun to play, aesthetically appealing to look at, and something users would want to play again.

What: For the game to work, players need to be able to choose a location for their next move, see the game board, know whose turn it is, know when the game is over, see the top scores, and start/restart the game.

Who: The users are groups of two people. General population but probably skewed more towards younger people. Players will be at least about 5 years old or however old you need to be to have enough reasoning skills to understand how to play connect four. Other stakeholders are the people purchasing this game, the companies making similar multi-player arcade games, and parents of children playing the game

Where/when: I envision this system as a standalone arcade game maybe at like a Chuck-e-Cheese, a movie theater lobby or pizza restaurant. The environment will be chaotic, with lots of lights and noises coming from other games and people. If there's other games around it, the system will have to compete for the user's attention.

Why: Since this is a game, people are playing to have fun. Some users might be using this game competitively, whether they are trying to beat their opponent or get one of the top scores. People may have specifically chosen this game because they are familiar with Connect Four as a board game.

Personas:

Name: Bella

Background:

9, Female

In 3rd grade in elementary school

Has not played connect 4 before (in real life or virtual), and likes computer games like Minecraft

Motivations:

Being constantly entertained

Trying to beat her older brother in some game

Frustrations:

Being told what to do by her parents instead of getting to do her own thing

Having to focus for more than 1-2 minutes on something unexciting

Bio:

Bella is a 9-year-old 3rd grader. Her favorite things to do afterschool are playing dress up, watching cartoons, and going to soccer practice. She is a fun, active youngster who's always looking for something new to entertain her. If Bella isn't nearly immediately captivated, she's quick to lose focus and will switch her attention to something different. Since she is a child in the digital age, she doesn't have much experience with physical board games and has never seen connect 4 before. However, she's well versed in computer games like Minecraft. Bella has been exposed to technology nearly all her life and even had to use it for learning during the COVID lockdowns. She does not like being told what to do. Some might think she's a brat, but really, she is just a blossoming independent woman.

Name: Thomas

Background:

45, Male

Manager at Office Depot

Married with 2 kids: both girls

Motivations:

Keep his children happy

Make the trip to the arcade short and uneventful so he can get home and destress from his job

Frustrations:

Getting out smarter by his young children while using technology



Having to play game to keep his kids entertained that are excruciatingly boring to adults

Changing technology. It seems that everything he uses does not meet his expectations

Bio:

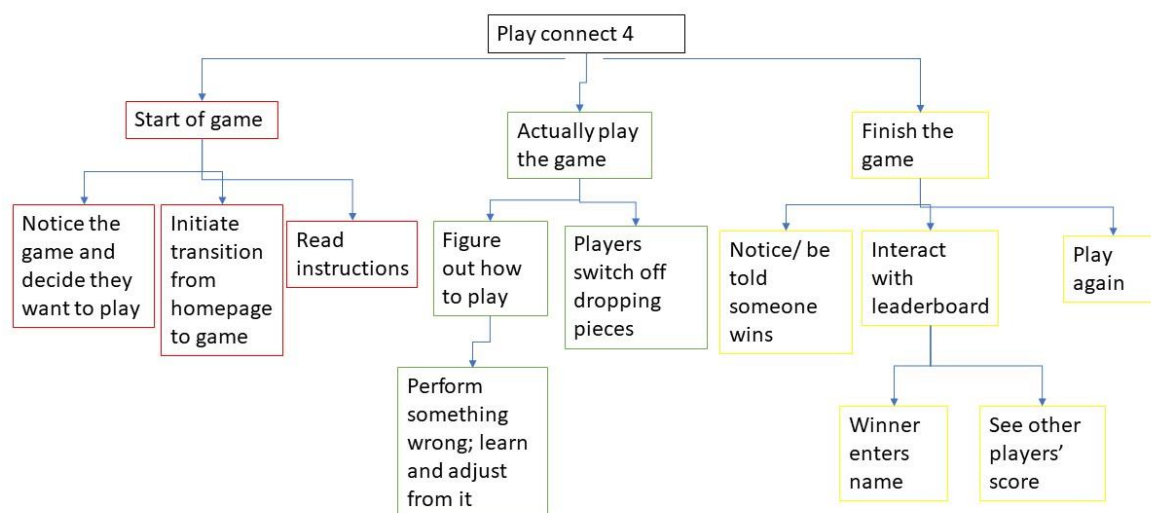
Thomas is a 42-year-old dad of two and manager at Office Depot. His hobbies include beer tasting, comic book collecting, and origami. He fondly remembers playing connect 4 as a child and mostly likely could recall all the rules of the simple game.

He tries his best to be a good dad, but it can be hard. Thomas can get overwhelmed trying to deal with the stress of keeping his kids happy on top of his stress from work. It is also not easy feeling like your young children are smarter than you when they seamlessly operate any tech they come across. For certain things, Tom prefers the “old fashion way” even if there’s more efficient ways since he doesn’t want to dedicate the time to learning fancy gadgets.

Use Case:

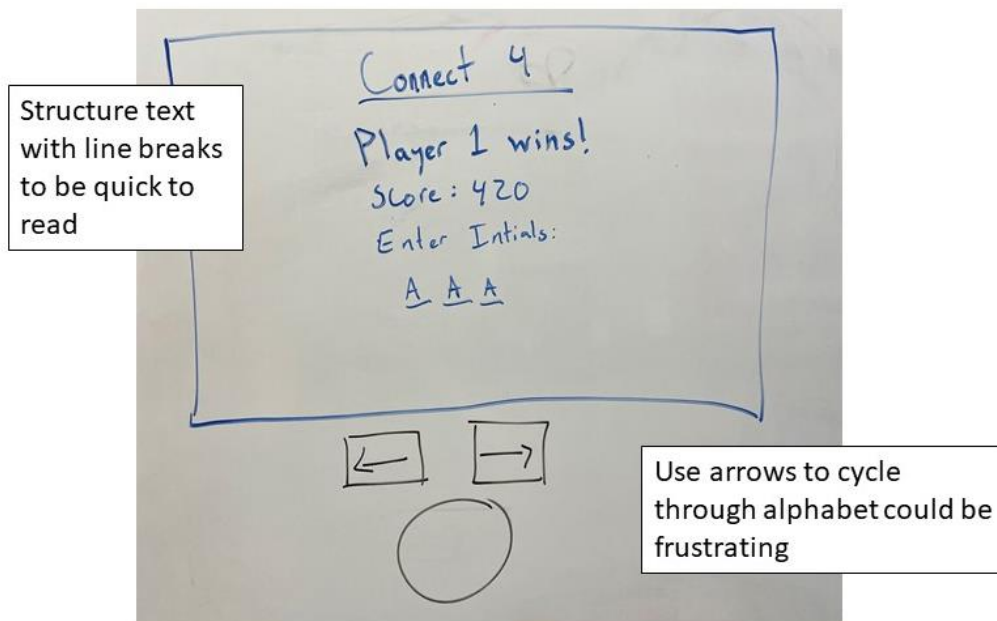
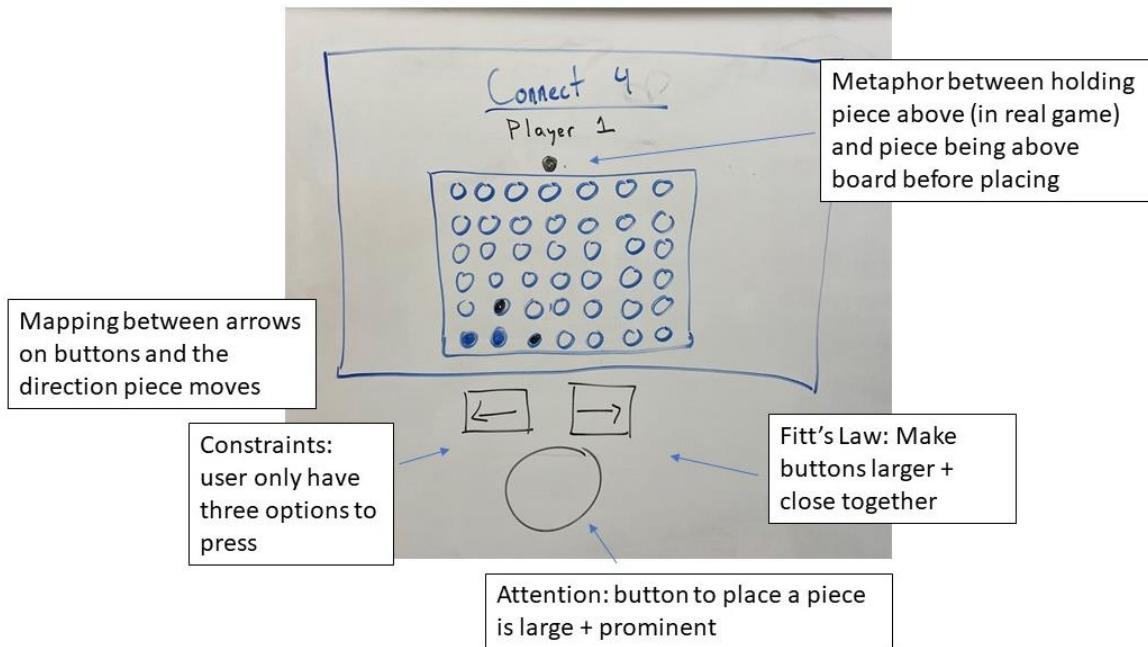
- System prompts the user to begin a game
- User chooses to start a game
- System displays instructions to user
- System displays the board and asks for the first player to make their move
- Users take turns picking a desired location for their next move
- After each turn, the system updates display to reflect the state of board with the new piece
- After each turn, the system prompts the next player to make their move
- Eventually, one of the users get four pieces in a row
- System indicates which player won and their score
- System prompts winning user to enter their name
- User enters their name
- System displays an updated version of the leader board
- System returns to the start page, prompting the user to begin a game

The below hierarchy describes how users might go about using the Connect 4 machine.

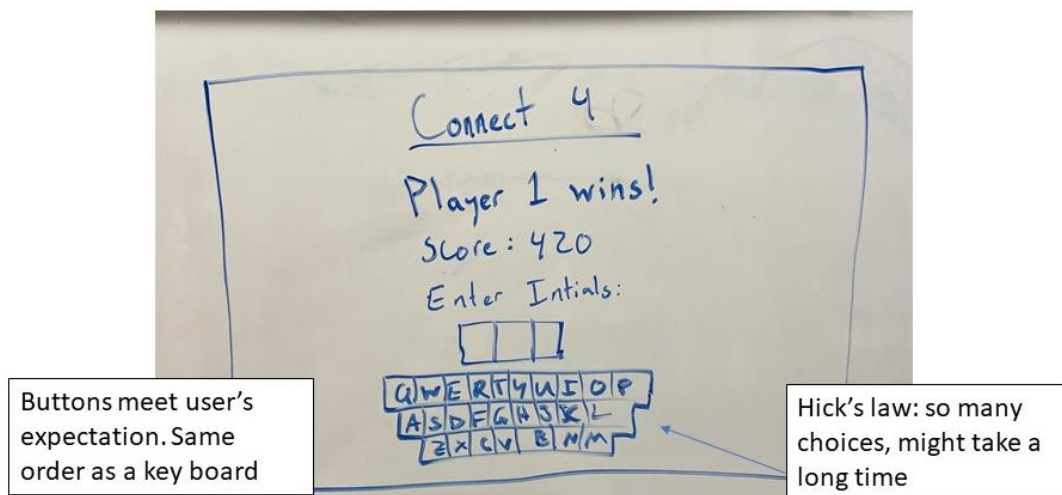
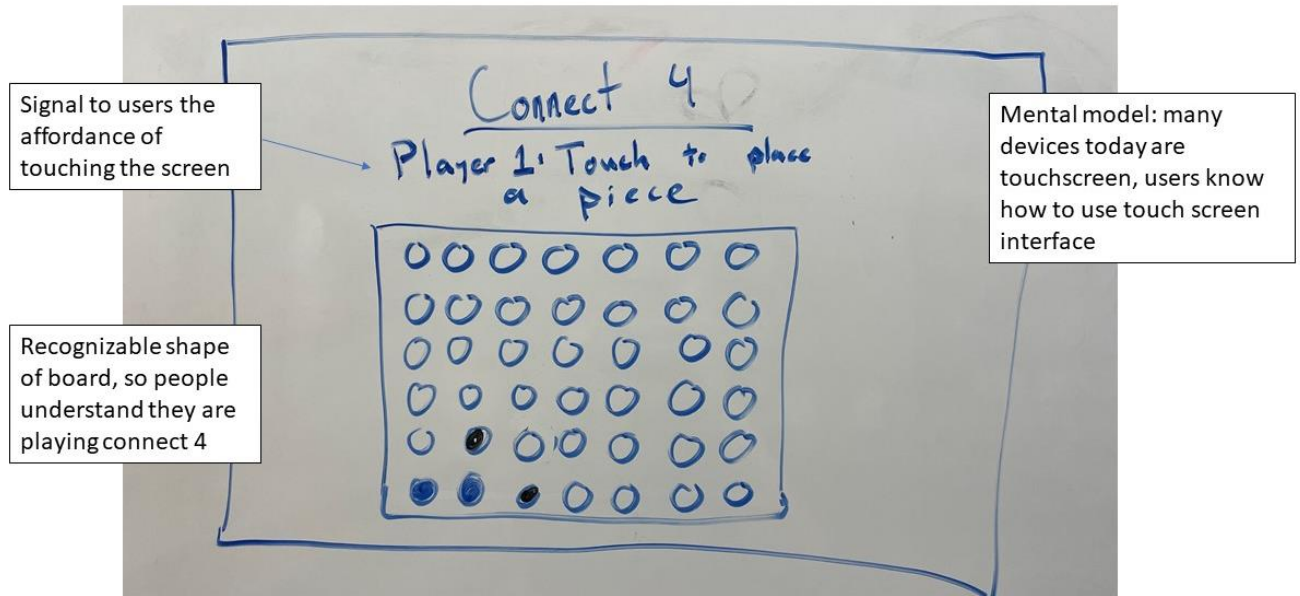


Design:

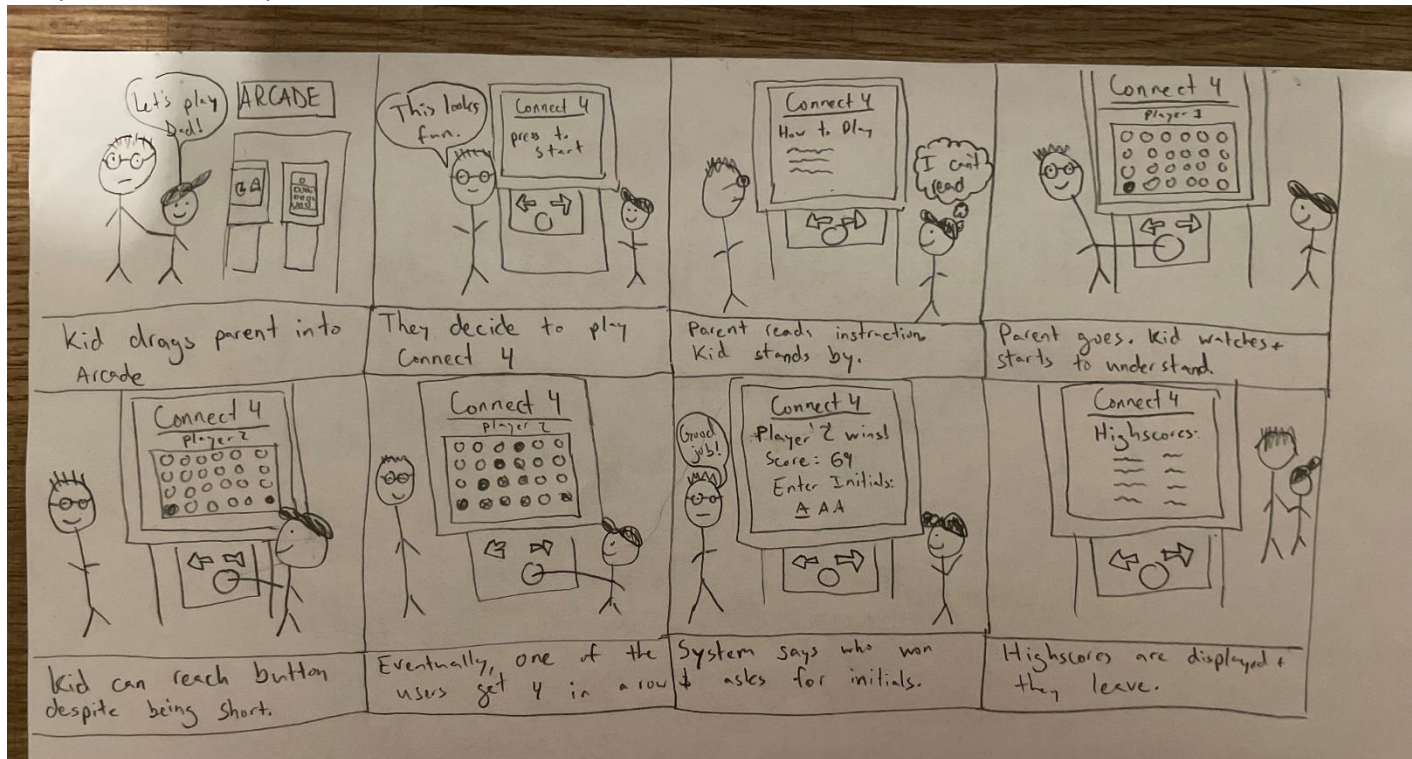
Concept 1



Concept 2



Story board for concept 1

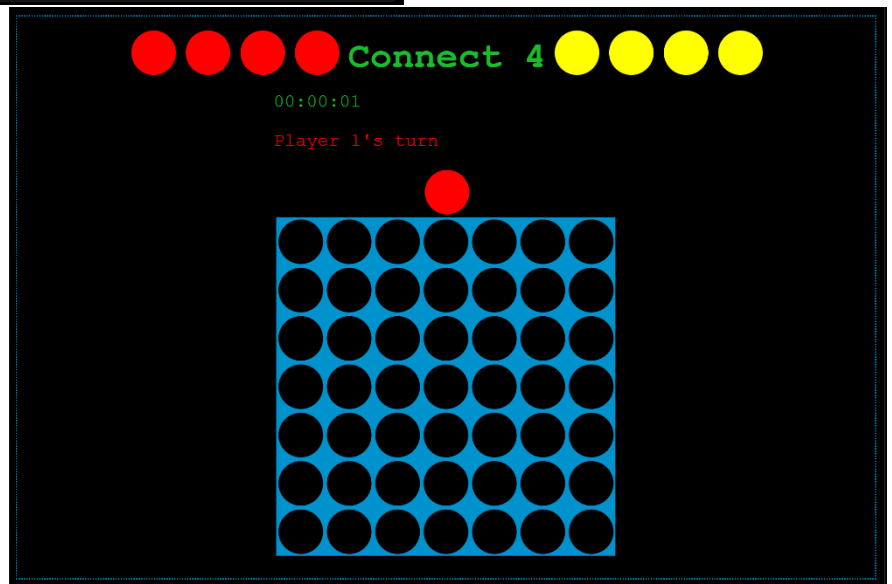
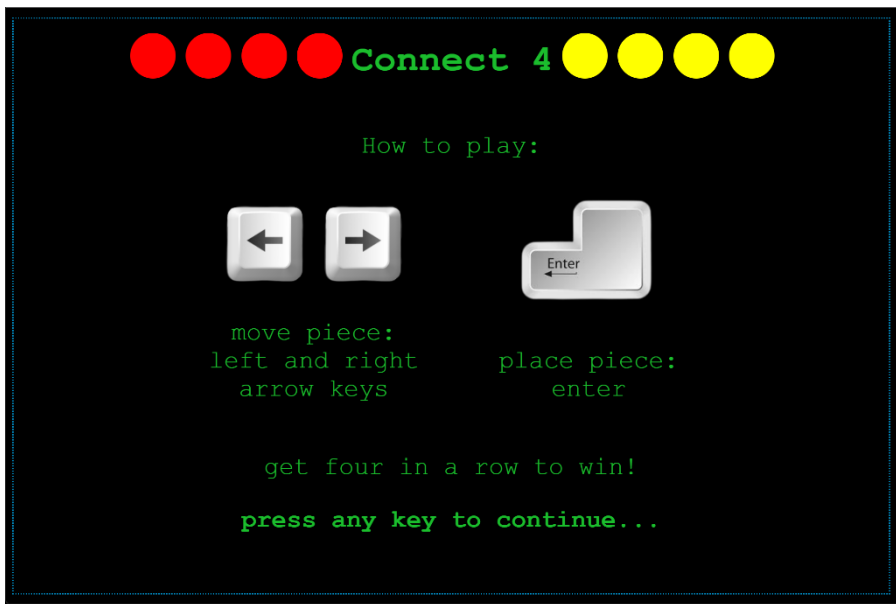
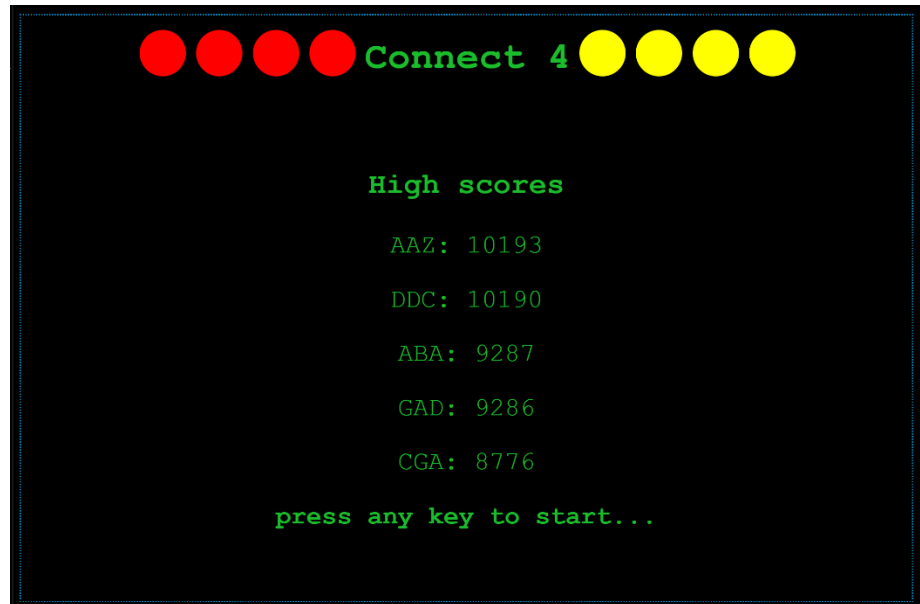


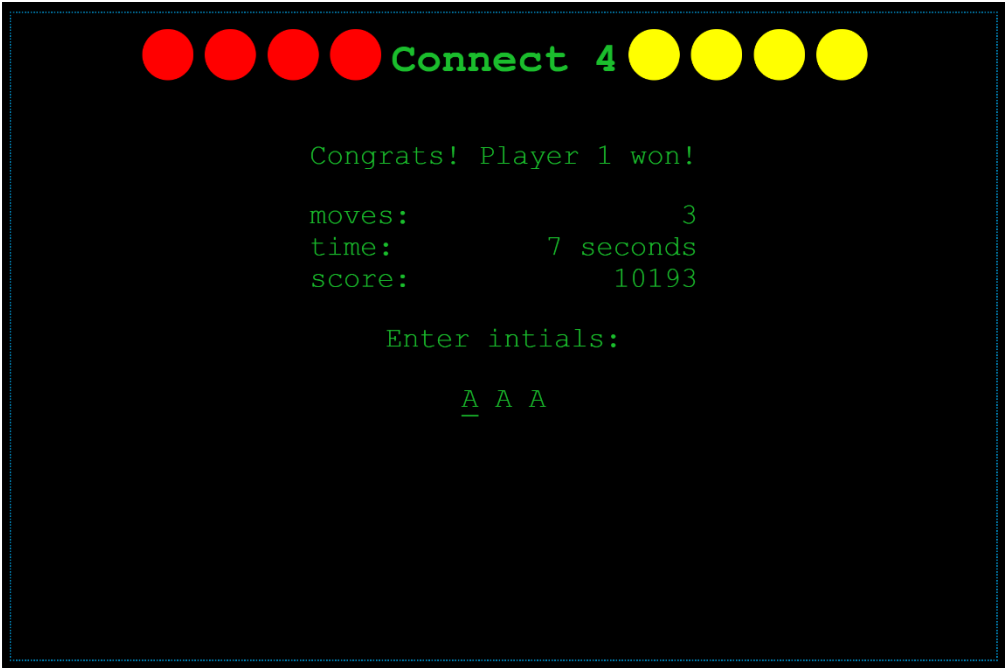
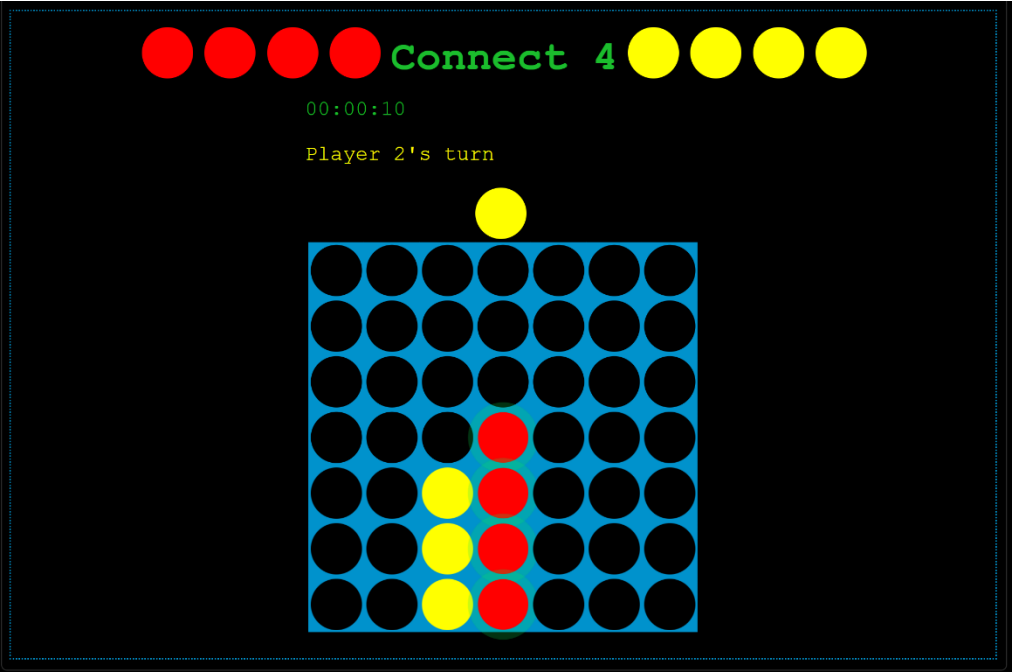
I have decided to go with the first design concept over the second because I see more benefits with a button interface than the touch screen interface. Although being able to touch the screen is commonplace in many other devices in today's world, there are many potential issues. Touch screen has no constraint on where the user can press. There could be difficulties touching the exact right spot. Someone might miss-touch slightly and the piece will not go in the column they intended. This is not an issue with the buttons because the user sees the piece above the row before confirming to drop it. The touch screen may be less accessible to certain users. For example, someone very short would not be able to reach the whole screen. The screen to enter initials feels crowded with a button for each letter. Additionally, as Hick's Law says, all the options increase the time it will take the user to make a choice.

The design will include aspects that are like the original connect four so that the users can recognize them and then remember how the game works (if they have previously played connect 4). The board will be the same shape the original. The piece about to be placed will be shown above the board, as if the user is holding and about to drop it.

Choice of colors will be an important part of the design. It is important that there is high contrast between the background color and colors of the elements on screen. This will ensure that the user can clearly see the game. The chosen colors also need to be fun and appealing. A user should be excited and drawn in, so they want to play the game and have fun. It will be best to use bright, happy colors instead of dull, sad colors.

Implementation:





Testing:

To evaluate my product, I let friends and family play through the game without me interfering. I first gave them a debrief by telling them this was a game for a class project and that they should try to play through it. As they played, I observed and then asked them the following questions. I ended up having 6 different people test out the game, and a summary of the responses is shown after each question.

1. How difficult was it to understand how to play (1 to 5) (Very difficult, difficult, neither, easy, very easy) Average: 3.8
2. How would you rate the experience of playing the game (1 to 5) (Very boring, boring, neither, fun, very fun) Average: 4
3. How would you rate the interaction of the game (1 to 5) (Very annoying, annoying, neither, pleasing, very pleasing) Average: 3.6
4. What did you think of the design of the game (1 to 5) (Very ugly, ugly, neither, attractive, very attractive) Average 3.6
5. Would you play this game again (1 to 5) (Definitely no, no, neither, yes, definitely yes) Average: 4.8
6. What did you like about the game? [fast paced, the color contrast, pictures in the instructions, timer]
7. What did you not like about the game? [Can't play with one person, wasn't obvious who won, need graphics after someone wins, not enough opportunity to customize]

While observing, I tried to keep track of any times someone did something "wrong" when interacting. All the users were able to operate the game without having issues. No one had to ask me questions on how to play or hesitated while trying to play. Two of the people struggled when entering their initials; they tried to use the keyboard to type instead of the arrow keys to scroll through the letters. I believe this issue is because this is only a prototype. The final product would be a machine with just three buttons and no keyboard, so I would expect more users to understand how to correctly enter initials in that case.

I also took note of the feedback received during the in-class demo. I see this almost as a heuristic evolution from experts, since everyone given feedback has taken this HCI class. The feedback I got from the class was to-be-dropped piece needs to be moved up, the blue is hard to see on black background, there should be a stall between the fourth piece being placed and the win screen, animations and/or sounds could be added, and the leader board should be on the start screen.

After testing, I took the suggestions and changed my program to reflect them, so the version that the class and my friends/family were looking at is different from what I'm turning in. I shifted up the to-be-dropped piece a few pixels, altered the board color to a different shade of blue, added an animation to highlight the four in a row after a win and combined the leader board and start screen into one page.